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Complex Networks

A1. Structure and Models

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Introduction

The study of complex networks provides a mathematical framework for understanding the organized complexity inherent in biological, social, and technological systems. By representing these systems as graphs—composed of nodes and their interconnecting edges—we can move beyond simple visual inspection to a rigorous quantitative characterization of their topology.

This report documents the structural analysis and modeling of five distinct networks provided in the **A1. Structure and Models** activity. The work is divided into two primary objectives:

1.1 Structural Characterization

In the first phase, we analyze four networks (`net1` through `net4`) to determine their macroscopic and microscopic properties. Macroscopic analysis involves calculating fundamental descriptors such as degree statistics, clustering coefficients, assortativity, and path lengths. Furthermore, we examine the degree distributions using linear and log-log scales to identify the underlying growth mechanisms of the networks.

At the microscopic level, we evaluate node importance through three distinct centrality metrics: **degree**, **betweenness**, and **eigenvector centrality**. This comparison allows us to determine whether different definitions of "relevance" converge on the same set of influential nodes.

1.2 Synthetic Model Identification

The second phase focuses on "reverse-engineering" the generation process of the networks. Based on the calculated descriptors, we classify `net1`–`net4` into one of four fundamental network models:

- **Erdős–Rényi (ER):** Randomly connected graphs.
- **Watts–Strogatz (WS):** Small-world networks with high clustering.
- **Barabási–Albert (BA):** Scale-free networks governed by preferential attachment.
- **Configuration Model (CM):** Assuming a power-law distribution with $\gamma < 2.5$.

Finally, the report investigates `net5`, an unknown spatial model. By visualizing the network using provided spatial coordinates and analyzing its connectivity and small-world properties, we propose a heuristic algorithm that accounts for its unique geographical distribution.

Part 1: Structural Characterization of Networks

This section details the analysis of four distinct networks (`net1` through `net4`) by examining both their global topology and local importance. While a vast array of parameters exists to describe complex systems, focusing on a strategic subset of macroscopic descriptors is often sufficient to capture the underlying structure of a given network. By quantifying these properties, we can effectively classify the networks into fundamental categories—such as random, small-world, or scale-free architectures—and evaluate the specific role of individual nodes through their centrality metrics.

2.1 Macroscopic Network Properties

To perform a comprehensive structural assessment of the provided networks, we extracted a set of global indicators. We quantified the network scale through node and edge counts, while degree statistics and clustering coefficients were used to evaluate local connectivity. Furthermore, we calculated assortativity, diameter, and average path length to characterize the overall topological framework and information-flow efficiency of each system.

Table 2.1: Summary of Global Network Characteristics

Property	**net1**	**net2**	**net3**	**net4**
Nodes	5000	5000	5000	5000
Edges	25000	24873	23508	24975
Min Degree	6	1	3	5
Max Degree	16	24	732	210
Mean Degree	10.00	9.9492	9.4032	9.99
Clustering Coeff.	0.4141	0.0021	0.0862	0.0107
Assortativity	-0.0097	-0.0057	-0.1339	-0.0325
Avg. Path Length	5.1211	3.9560	3.0082	3.4868
Network Diameter	8	7	5	5

2.1.1 Initial Observations

A comparative analysis of the metrics in Table 2.1 highlights the fundamental topological differences between the four datasets. While they share a common scale ($N = 5000$), their structural properties suggest divergent formation mechanisms:

- **Connectivity Extremes:** The degree distribution range acts as a primary differentiator. **Net1** is characterized by a highly homogeneous structure (Max Degree 16), whereas **net3** and **net4** exhibit significant heterogeneity. Specifically, **net3** possesses a massive hub ($k_{max} = 732$), a hallmark of scale-free architectures where a few nodes command a disproportionate number of links.

- **Local Transitivity:** The clustering coefficient varies by two orders of magnitude across the samples. **Net1** shows a high density of triangles ($C = 0.4141$), indicating a robust local community structure. Conversely, the near-zero clustering of **net2** (0.0021) suggests a lack of local organization, typical of purely random graphs where edges are independent of their neighbors' positions.

- **Global Efficiency and Path Metrics:** Despite its lower edge count, **net3** is the most "compact" system, with the lowest average path length (3.0082) and diameter (5). This demonstrates the "Small-World" efficiency provided by high-degree hubs, which act as topological bridges to minimize the distance between any two nodes in the system.

- **Mixing Patterns:** All four networks display negative assortativity, most notably **net3** (-0.1339). This disassortative nature implies a "star-like" connectivity preference where influential hubs tend to connect to low-degree peripheral nodes, a common feature in technological and biological networks that enhances system-wide stability.

These initial findings indicate that the networks do not share a common generative origin. The high clustering of **net1** contrasts sharply with the hub-centric nature of **net3** and the randomness of **net2**. In the sections that follow, we perform a detailed examination of the degree distributions and additional structural features to better define these differences. Analyzing the distribution of connections is essential, as it reveals the organizational patterns that allow us to accurately classify each network type.

2.1.2 Degree Distribution Analysis

The degree distribution, $P(k)$, provides a visual profile of how links are distributed across the nodes in a network. By examining this property through both linear and log-log scales, we can identify specific patterns in connectivity. Specifically, these plots help us determine if a network is homogeneous, where most nodes have a similar number of links, or heterogeneous, where a few highly connected "hubs" dominate the structure.

Network 1 (net1)

The structural analysis of **net1** reveals a highly uniform and homogeneous topology. As seen in the **linear scale histogram** (Figure 2.1, left), the degree distribution is tightly centered around the mean value ($\langle k \rangle = 10$), with the vast majority of nodes falling within a narrow range between a minimum degree of 6 and a maximum of 16. This high concentration around the mean indicates a "regularized" connectivity where no single node acts as a dominant hub.

In the **log-log visualization** (Figure 2.1, right), the data points form a distinct parabolic arc rather than the straight line characteristic of power-law systems. This

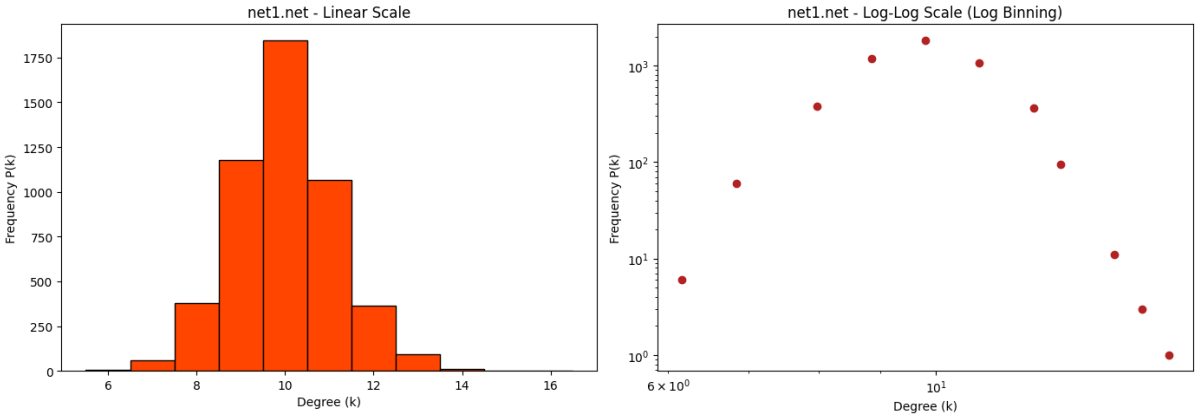

Figure 2.1: Degree distribution for net1 in linear (left) and log-log (right) scales.

downward curvature confirms the absence of scale-free properties and shows that the probability of finding a node decreases rapidly as the degree increases beyond the average.

The macroscopic metrics further characterize this balanced structure. Despite the large size of the network (5,000 nodes), the average path length remains relatively short (5.1211) with a diameter of 8. Notably, the clustering coefficient is quite high (0.4141), pointing to a strong presence of local triangles and community-like connectivity. Finally, the near-zero assortativity (-0.0097) indicates that node connections are not significantly influenced by the degree of their neighbors, resulting in a stable and non-hierarchical network layout.

In summary, **net1** is a homogeneous network with a very narrow degree range and no significant hubs. Its well-balanced structure suggests that its connectivity is driven by localized or random processes rather than a preferential attachment mechanism.

Network 2 (net2)

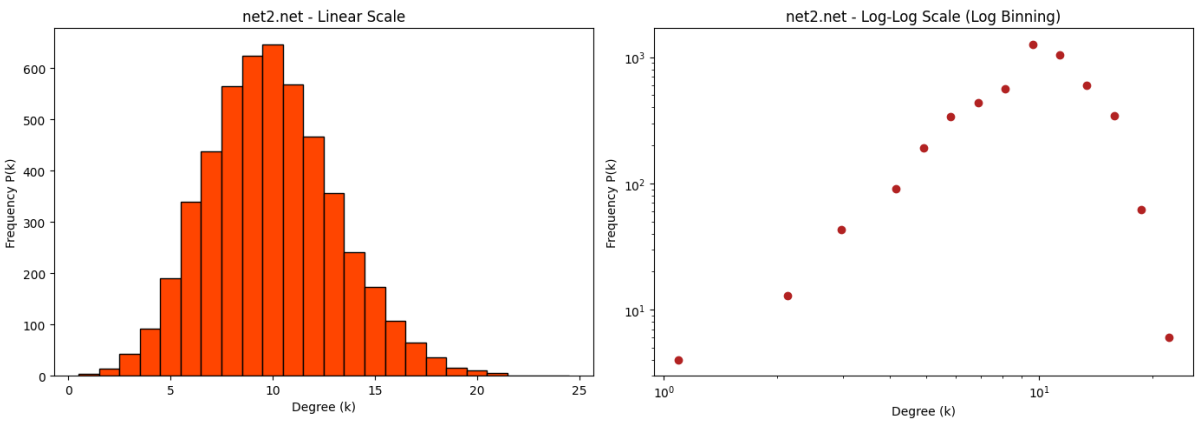

Figure 2.2: Degree distribution for net1 in linear (left) and log-log (right) scales.

The connectivity of **net2** shows a homogeneous structure that is similar to **net1** in its average degree ($\langle k \rangle \approx 9.95$), but with a wider spread of values. As seen in the **linear histogram** (Figure 2.2, left), the distribution follows a bell-shaped curve. However, it is more spread out than **net1** , with degrees ranging from 1 to 24. While this shows more

variety in how many links a node can have, the network still does not have any extremely high-degree nodes (hubs).

In the **log-log plot** (Figure 2.2, right), there is no straight line, which proves that the network does not follow a power-law distribution. Instead, the curve drops off quickly at higher degrees. This confirms that **net2** is not a scale-free network and that the chance of finding a node with many connections is very low.

A major difference in this network is its very low clustering coefficient (0.0021), which is much smaller than the clustering found in **net1**. This means that the neighbors of a node are almost never connected to each other, showing a lack of local "communities." Additionally, the average path length is shorter than in **net1** (3.9560 vs 5.1211), suggesting a more "random" and spread-out structure where nodes are easily reached but not tightly grouped.

In summary, **net2** has a more dispersed but still homogeneous degree distribution compared to **net1**. Its very low clustering and shorter path lengths suggest a structure where links are placed more randomly, without forming the tight local groups seen in the previous network.

Network 3 (net3)

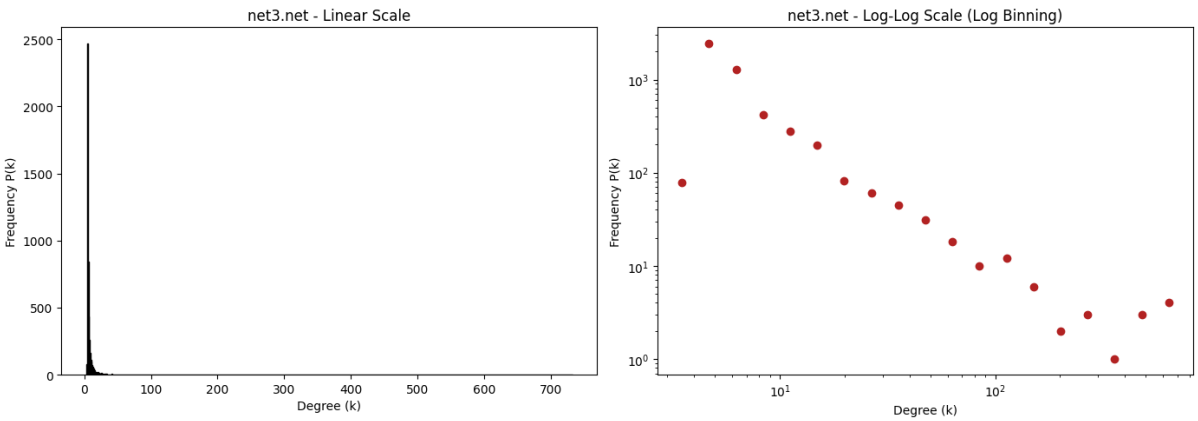

Figure 2.3: Degree distribution for net1 in linear (left) and log-log (right) scales.

The connectivity of **net3** marks a clear change from the more even patterns seen in the first two networks. In the **linear scale** (Figure 2.3, left), the histogram shows that most nodes have very few connections, but there is a long "tail" reaching a maximum degree of 732. This shows that while most nodes are small, the network contains massive hubs that hold the structure together.

The **log-log plot** (Figure 2.3, right) provides the strongest evidence for the nature of this network. Unlike the curved lines of the previous networks, the points here follow a remarkably straight linear trend. This "straight-line" behavior on a logarithmic scale is the classic signature of a power-law distribution, which indicates a scale-free architecture.

These observations are backed up by the global metrics. The presence of large hubs significantly reduces the distance between nodes, resulting in the shortest average path length (3.0082) and the smallest diameter (5) of all the networks. Additionally, **net3** shows a notable negative assortativity (-0.1339). This suggests a "hub-and-spoke" style of connectivity, where these highly connected hubs tend to link with small, low-degree nodes.

In conclusion, **net3** is a scale-free network. Its structure is defined by dominant hubs that increase the network's overall connectivity and efficiency much more than the uniform layouts of **net1** and **net2**.

Network 4 (net4)

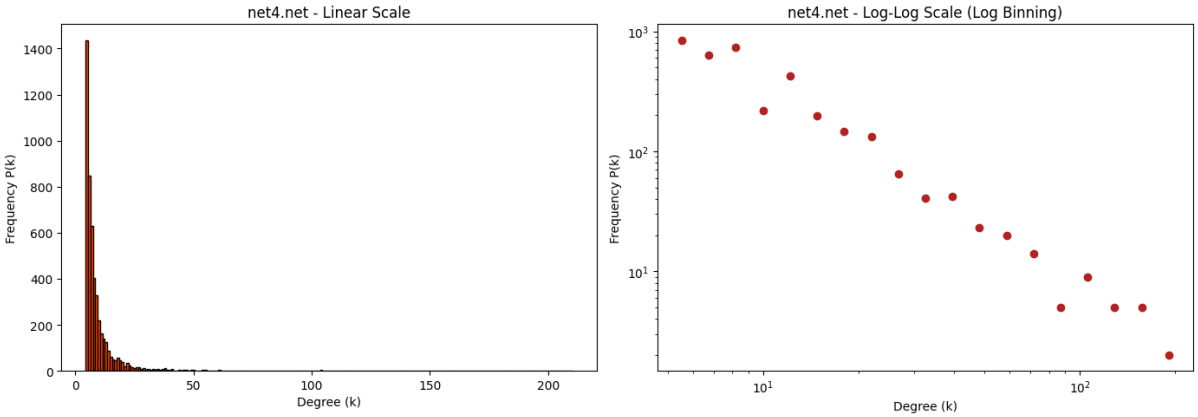

Figure 2.4: Degree distribution for net1 in linear (left) and log-log (right) scales.

The distribution of **net4** exhibits several characteristics similar to **net3**, pointing toward a scale-free architecture. In the **linear histogram** (Figure 2.4, left), the network shows a clear heavy-tailed pattern with a maximum degree of 210. While these hubs are not as large as those in **net3**, they still represent a significant departure from the more uniform distributions of the first two networks.

In the **log-log plot** (Figure 2.4, right), the data shows a strong downward trend. Although there is more curvature compared to the distribution of **net3**, the presence of high-degree nodes across several orders of magnitude suggests that **net4** follows a heavy-tailed distribution typical of scale-free systems.

The global metrics reinforce this classification. The network is highly efficient, with an average path length of 3.4868 and a diameter of 5, which are common traits in networks where hubs facilitate quick navigation. Similar to **net2**, the clustering coefficient is low (0.0107), indicating that the network relies more on its global hub structure than on local community grouping. The slightly negative assortativity (-0.0325) further suggests a structure where these hubs connect to a variety of lower-degree nodes.

Overall, **net4** is a heterogeneous network with scale-free characteristics. Its structure is defined by the presence of significant hubs that provide high connectivity and efficiency, placing it in a similar category to **net3** despite having a different degree range.

2.2 Microscopic Structure Analysis

In this section, we shift from global network properties to the analysis of individual node importance. We evaluate the microscopic structure by calculating three primary centrality metrics: **Degree Centrality**, **Betweenness Centrality**, and **Eigenvector Centrality**. Table 2.2 summarizes the five most influential nodes for each network according to these indicators.

Table 2.2: Top 5 nodes by centrality. Format: **Node ID (Value)**

Net	Degree Rank	Degree Node (Val)	Betweenness Rank	Betweenness Node (Val)	Eigenvector Rank	Eigenvector Node (Val)
net1	1	1693 (16)	1	4747 (0.00406)	1	651 (0.02863)
	2	651 (15)	2	2645 (0.00405)	2	1937 (0.02647)
	3	1579 (15)	3	230 (0.00380)	3	4526 (0.02563)
	4	4891 (15)	4	4360 (0.00379)	4	4398 (0.02525)
	5	41 (14)	5	1579 (0.00366)	5	1939 (0.02517)
net2	1	1581 (24)	1	1581 (0.00330)	1	1581 (0.04122)
	2	787 (23)	2	787 (0.00256)	2	3233 (0.03434)
	3	52 (21)	3	4382 (0.00246)	3	787 (0.03413)
	4	1990 (21)	4	52 (0.00227)	4	2375 (0.03374)
	5	2372 (21)	5	2375 (0.00227)	5	131 (0.03355)
net3	1	5 (732)	1	5 (0.13781)	1	5 (0.25923)
	2	7 (691)	2	7 (0.12767)	2	7 (0.24168)
	3	2 (626)	3	0 (0.11134)	3	2 (0.22540)
	4	0 (620)	4	2 (0.10930)	4	0 (0.22536)
	5	6 (555)	5	6 (0.09429)	5	3 (0.20698)
net4	1	6 (210)	1	0 (0.06050)	1	6 (0.22391)
	2	0 (206)	2	6 (0.05759)	2	0 (0.22342)
	3	9 (204)	3	9 (0.05526)	3	9 (0.20618)
	4	10 (163)	4	8 (0.03907)	4	8 (0.17408)
	5	8 (162)	5	3 (0.03793)	5	3 (0.16858)

2.2.1 Comparison of Centrality Indicators

Comparing the metrics reveals whether different indicators provide redundant or complementary information about node relevance.

In **net1**, there is a significant discrepancy between the three rankings. The node with the highest degree (ID 1693) does not appear in the top lists for betweenness or eigenvector centrality. Similarly, node 4747 is the most important for information flow (Betweenness) but is not among the most connected. This suggests that in this network, node importance is multi-faceted: a node can be a "bridge" without being "popular," and having many links does not necessarily imply a central position in the global structure.

In contrast, **net3** and **net4** show a high degree of correlation between the indicators. In **net3**, nodes 5, 7, 0, and 2 occupy the top positions across all three metrics. This convergence is typical of scale-free networks, where massive hubs naturally centralize the network; they act as the primary bridges for shortest paths (High Betweenness) and are connected to other influential nodes (High Eigenvector).

net2 presents an intermediate case. While node 1581 leads in all categories, the other rankings vary slightly. However, the metrics are more aligned than in **net1**, likely due to the higher degree variance observed in the earlier distribution analysis.

Relevance of the Indicators

The centrality indicators do not always provide the same information. Their alignment is a property of the network topology itself:

- In **homogeneous networks** (**net1**, **net2**), the indicators provide distinct information. One must consider all metrics to understand the different roles (bridges vs. hubs) nodes play.
- In **heterogeneous networks** (**net3**, **net4**), the indicators are largely redundant. The hubs are so dominant that they become the most relevant nodes regardless of the metric used.

The absolute values also provide insight: the betweenness values in **net3** (up to 0.137) are much higher than in **net1** (approx. 0.004). This confirms that the hubs in **net3** have a much stronger "monopoly" over the network's communication paths than any node in the first two networks.

2.3 Correlation of Centrality Metrics

To evaluate the consistency between different node importance measures, we calculated the Spearman rank correlation coefficients (ρ) for each network. This metric determines the extent to which different centrality indicators agree on the ranking of the nodes. The results are summarized in Table 2.3.

Table 2.3: Comparison of Spearman Rank Correlation Coefficients (ρ) across metrics.

Network	Degree / Betweenness	Degree / Eigenvector	Betweenness / Eigenvector
net1	0.6217	0.7157	0.4559
net2	0.9919	0.9436	0.9552
net3	0.6089	0.4651	0.7061
net4	0.8915	0.5628	0.7894

2.3.1 Interpretation of Correlations

The observed coefficients highlight fundamental differences in how these networks are organized:

- **Structural Redundancy in net2:** This network exhibits nearly perfect correlation across all metric pairs (> 0.94). This suggests a highly integrated structure where different measures of importance are redundant; a node's local connectivity (Degree) almost perfectly dictates its global influence (Betweenness) and its prestige (Eigenvector).
- **Specialized Roles in net1:** Conversely, **net1** shows the lowest correlations in the set, particularly between Betweenness and Eigenvector (0.4559). This implies that nodes have highly specialized functions. A node might be essential for routing information (high Betweenness) without necessarily being part of a high-status or densely connected cluster (low Eigenvector).
- **Variations in Hub Influence (net3 vs. net4):** While both networks exhibit heavy-tailed distributions, their internal hierarchies differ. **net4** shows a very strong link between Degree and Betweenness (0.8915), indicating that its hubs are the

primary "gatekeepers" of the network. In **net3**, this correlation is notably lower (0.6089), suggesting that some high-degree nodes are more localized and do not sit on a proportional number of shortest paths.

In conclusion, the data confirms that centrality indicators provide the most unique, non-overlapping information in **net1**. In contrast, for **net2** and **net4**, the high degree of interdependence suggests that the network's importance is concentrated such that a single metric is often sufficient to capture the overall node hierarchy.

2.4 Synthesis of Exploratory Network Analysis

The initial characterization of the four networks reveals two distinct structural paradigms: homogeneous systems with localized influence and heterogeneous systems dominated by global hubs.

Structural Archetypes Networks **net1** and **net2** represent relatively uniform topologies. **net1** is characterized by the highest average path length and the most significant functional differentiation among its nodes, as evidenced by the low correlation between centrality metrics. **net2**, while similar in its narrow degree distribution, displays a much higher degree of structural redundancy, where connectivity and central positioning are tightly coupled.

Hub-Dominated Hierarchies In contrast, **net3** and **net4** exhibit the hallmarks of scale-free architectures. Their degree distributions follow heavy-tailed patterns, resulting in the emergence of massive hubs. These hubs facilitate high global efficiency, leading to shorter average path lengths and smaller diameters compared to the homogeneous models. In these networks, node importance is concentrated, and different centrality measures tend to converge on the same set of dominant nodes, particularly in **net4**.

Table 2.4: Summary of structural properties across the four networks.

Property	**net1**	**net2**	**net3**	**net4**
Distribution	Homogeneous	Homogeneous	Heterogeneous	Heterogeneous
Connectivity	Regular-like	Random-like	Scale-free	Scale-free
Hub Impact	Negligible	Low	Extreme	Significant
Metric Correlation	Low	Very High	Moderate	High

The evidence gathered through these analyses provides a robust foundation for the formal model identification in the following chapter. Based on these observations, we can begin to categorize the networks according to established theoretical models such as Erdős-Rényi, Watts-Strogatz, Barabási-Albert, and Configuration Model.

Part 2: Models

3.1 Network Model Identification

In this section, we utilize the global metrics and microscopic data derived in the previous chapters to identify the generative models for the four analyzed networks. We consider four theoretical frameworks: the Erdős-Rényi (ER) model, the Watts-Strogatz (WS) model, the Barabási-Albert (BA) model, and the Configuration Model (CM).

3.1.1 Categorization and Justification

Based on the quantitative evidence, the networks are classified into four distinct generative models. The transition from homogeneous to heterogeneous structures is clearly reflected in the interplay between the clustering coefficient (C), the average path length (L), and the power-law exponent (γ).

Table 3.1: Empirical evidence and model identification for net1–net4.

Network	$\langle C \rangle$	$\langle L \rangle$	**Max** K	**Est.** γ	**Identified Model**
net1	0.4141	5.1211	16	2.98	Watts-Strogatz (WS)
net2	0.0021	3.9560	24	1.45	Erdős-Rényi (ER)
net3	0.0862	3.0082	732	2.24	Configuration Model (CM)
net4	0.0107	3.4868	210	3.12	Barabási-Albert (BA)

- **net1: Watts-Strogatz (WS) Model**
  This network is identified as a WS model with a low-to-intermediate rewiring probability. The primary evidence is the high clustering coefficient ($C = 0.4141$), which is several orders of magnitude higher than a random graph of the same density ($C_{rand} \approx 0.002$). Despite this high local order, the average path length remains low ($L = 5.12$), a classic signature of the "Small-World" phenomenon where local neighborhoods are preserved while long-range shortcuts reduce global distance.

- **net2: Erdős-Rényi (ER) Model**
  net2 follows the ER random graph model. Its clustering coefficient is nearly zero ($C = 0.0021$), aligning with the theoretical expectation for random edges ($p \approx \langle k \rangle / N$). Furthermore, it exhibits the typical "random-like" properties discussed earlier: a very short path length ($L = 3.95$) and a degree distribution that lacks the heavy tail seen in hub-dominated systems (Max $K = 24$).

- **net3: Configuration Model (CM)**
  This network is identified as a Configuration Model with a heavy-tailed degree

distribution where $\gamma < 2.5$. The estimated $\gamma = 2.24$ places it in the "ultra-small world" regime. This is corroborated by the existence of massive hubs (Max $K = 732$), which account for nearly 15% of the total node connectivity. In this regime, the average path length L is expected to scale as $\ln(\ln N)$, explaining why **net3** has the shortest distance of all four networks ($L = 3.0082$).

- **net4: Barabási-Albert (BA) Model**
  net4 is identified as a BA model generated via preferential attachment. The degree distribution yields a power-law exponent of $\gamma = 3.12$, which is consistent with the theoretical prediction of $\gamma = 3$ for the standard BA model. While it is scale-free, its hubs are significantly smaller than those in **net3** (Max $K = 210$ vs. 732), indicating a more balanced growth process compared to the extreme heterogeneity of the Configuration Model.

3.2 Analysis of net5: A Non-Standard Model

The final stage of this study involves the characterization of **net5**, a network generated by a model not covered in the theoretical lectures. By integrating topological metrics with the spatial data from `positions_net5.txt`, we can identify the underlying generative principles of this system.

3.2.1 Visual and Structural Properties

The visual representation of **net5** (see Figure 3.1) reveals a structure fundamentally different from the previous abstract models. The most prominent feature is the **spatial dependency** of its links: nodes are exclusively connected to their immediate physical neighbors, creating a "patchy" topology with localized clusters.

To quantitatively describe these features, the key structural metrics are summarized in Table 3.2.

Table 3.2: Structural metrics for net5 and its Largest Connected Component (LCC).

Metric	**Value**
Number of nodes	200
Number of edges	465
Number of components	7
Minimum degree	1
Maximum degree	10
Average degree	4.6500
Avg Clustering Coeff	0.6107
Assortativity	0.5502
Avg Path Length (LCC)	13.1310
Diameter (LCC)	35

The structural analysis provides the following evidence:

- **Connectivity:** The network is disconnected, consisting of 7 distinct components. This is typical of spatial models where nodes in low-density areas may fall outside the connection range of the rest of the system.

- **Scale-Free Property:** The network is not scale-free. The degree distribution (Figure 3.2) shows a clear peak around the average degree ($\langle k \rangle = 4.65$) rather than a power-law decay. The maximum degree is quite low ($k_{max} = 10$), indicating the absence of global hubs.

- **Small-World Property:** The Largest Connected Component (LCC) does not exhibit small-world behavior. While the clustering coefficient is remarkably high ($C = 0.6107$), suggesting dense local neighborhoods, the average path length is very large ($L = 13.1310$) with a diameter of 35. In the absence of long-range "shortcuts," information must travel hop-by-hop across the physical space.

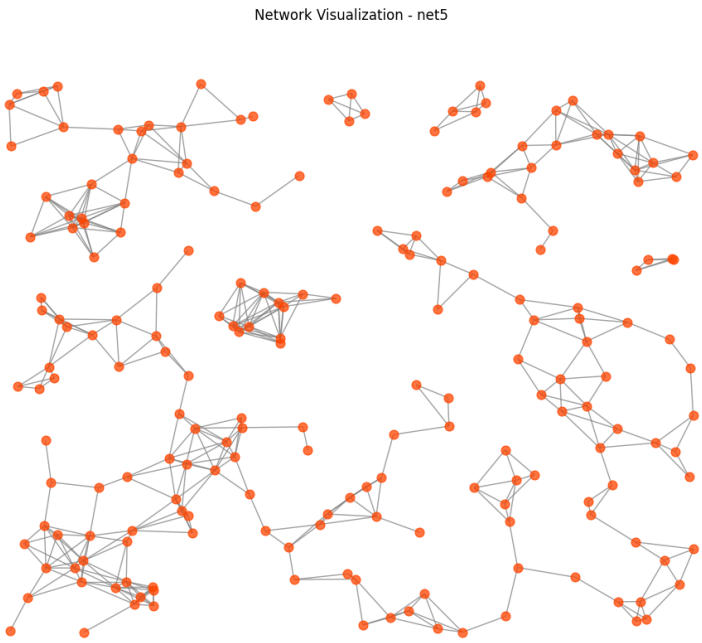

Figure 3.1: Spatial visualization of net5 using provided coordinates.

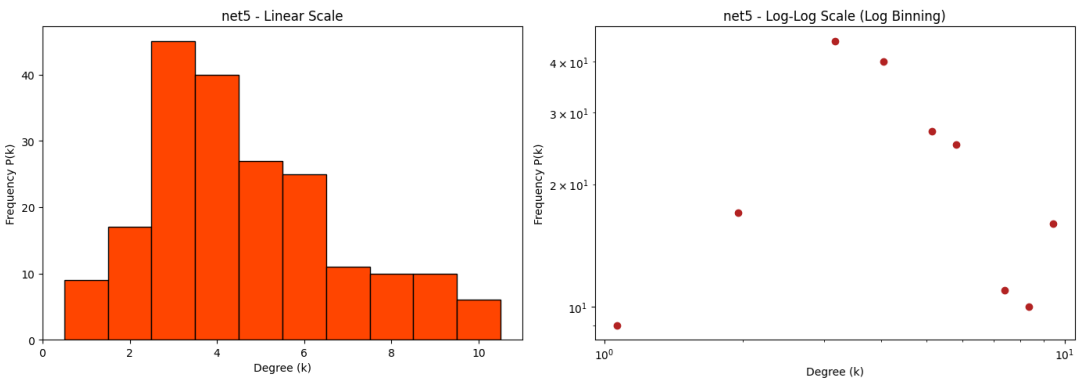

Figure 3.2: Degree distribution for net5 in linear (left) and log-log (right) scales.

3.2.2 Proposed Generative Algorithm

Based on the empirical evidence that connectivity in **net5** is strictly a function of Euclidean proximity, we identify this system as a **Random Geometric Graph (RGG)**. This model is fundamentally different from abstract topological models like Erdős-Rényi or Barabási-Albert, as it assumes that nodes exist in a metric space where distance determines the probability of interaction. The proposed algorithm for generating such a network is as follows:

1. **Spatial Distribution:** Distribute $N = 200$ nodes randomly within a two-dimensional metric space (typically a unit square $[0, 1]^2$). Each node i is assigned a position vector $\vec{x}_i = (x_i, y_i)$ sampled from a uniform probability distribution.

2. **Distance Thresholding:** Define a critical radius $r \in [0, \sqrt{2}]$, representing the maximum communication range or interaction distance. This parameter is the primary determinant of the network's density and global connectivity.

3. **Edge Creation:** For every unique pair of nodes (u, v), calculate the Euclidean distance:
$$d(u, v) = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2}$$
An undirected edge is established if and only if $d(u, v) \leq r$.

Justification of the Model Selection

The RGG model is the most scientifically sound choice for **net5** because it simultaneously accounts for all four observed structural anomalies that standard models fail to explain:

- **Local Clustering vs. Global Inefficiency: net5** exhibits a very high clustering coefficient ($C = 0.6107$) alongside an exceptionally large diameter ($D = 35$) and average path length ($L = 13.13$). In an RGG, triangles are formed naturally because if nodes v and w are both within radius r of node u, they are physically likely to be within radius r of each other. Conversely, the lack of "long-range shortcuts" (edges connecting distant coordinates) forces all global communication to travel through long, local chains, explaining the high path length.

- **Structural Fragmentation:** The presence of 7 distinct components is a signature of a spatial graph operating near the percolation threshold. Unlike the Erdős-Rényi model, where a "giant component" usually absorbs isolated nodes quickly as edges increase, spatial gaps in an RGG leave nodes in low-density areas permanently isolated.

- **Bounded Degree Distribution:** The degree distribution is peaked and bounded, with a maximum degree of only $k_{max} = 10$. This is consistent with uniform node placement in a 2D plane; a node's degree is limited by the number of other nodes that can physically fit within the area defined by its radius πr^2. This eliminates the possibility of the Barabási-Albert (Scale-Free) model, which would require high-degree hubs that bypass spatial constraints.

- **Direct Spatial Alignment:** The visualization clearly demonstrates that edges do not cross each other over long distances. Every link in the network correlates

exactly with the proximity of coordinates provided in the `positions_net5.txt` file, making a distance-based criterion the only viable explanation for the observed topology.

Conclusion

This report provided a comprehensive structural analysis of five distinct networks, transitioning from exploratory characterization to formal model identification. By synthesizing global topological metrics, degree distribution analysis, and microscopic centrality correlations, we successfully categorized each system into its respective generative paradigm.

Identification of Synthetic Models The analysis of the first four networks revealed a clear dichotomy between homogeneous and heterogeneous architectures.

- **net1 and net2** represented the homogeneous regime, with **net1** exhibiting the "Small-World" characteristics of the **Watts-Strogatz** model (high clustering with global shortcuts) and **net2** following the purely stochastic nature of the **Erdős-Rényi** model.
- **net3 and net4** demonstrated the efficiency of scale-free structures. While both are dominated by hubs, the extreme heterogeneity and "ultra-small world" properties of **net3** pointed to a **Configuration Model** with a heavy-tail exponent ($\gamma < 2.5$), whereas **net4** aligned with the growth-based logic of the **Barabási-Albert** model.

Insights from Spatial Constraints The investigation into **net5** provided the most significant contrast. Unlike the previous models, which operate on abstract topological rules, **net5** was identified as a **Random Geometric Graph (RGG)**. The high clustering ($C = 0.6107$) and significantly larger path length ($L = 13.13$) observed in this network are direct consequences of physical distance acting as a constraint on connectivity. This analysis underscored that in real-world or spatially-embedded systems, the absence of long-range shortcuts leads to high local transitivity at the cost of global efficiency.

Final Synthesis Through this multi-scale approach, we have demonstrated that node importance is not a universal property but is deeply dictated by the generative mechanism of the network. High centrality correlations in random and scale-free models suggest structural redundancy, while low correlations in regular-like or spatial models highlight the emergence of specialized functional roles. Ultimately, the ability to match empirical data to theoretical models—such as ER, WS, BA, and RGG—provides a powerful predictive framework for understanding the resilience, efficiency, and evolution of complex systems.